

Nau mai, welcome to

Applied Engineering Optimisation

ENEL445

Semester One

New course

(2024 – special topic, 2025 onwards – permanent course)

- Why optimisation:
 - Optimisation is used a lot in industry and is one of the things industry have told us to teach more about.
 - Many hot topics (e.g. Artificial Intelligence / Machine Learning, Smart Grids, etc.) are based on optimisation.
 - If any of you are thinking about postgrad, the techniques taught in this course are likely to be useful to you.
- This is still a new course – please be aware that we don't have everything sorted and let us know if you find any issues.

Lecturers



Le Yang

Lecturer

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Office: Link Rm A510



Jeremy Watson

Course coordinator and lecturer

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Office: Link Rm A314

Visiting Lecturer



Prof. Roy Smith
Visiting Erskine Fellow

Lecture plan

- Week 1: Intro to optimisation, projects (Jeremy, Le)
- Weeks 2-3: Unconstrained optimisation, gradient-based descent, numerical gradient computation (Le)
- Weeks 4-6: Constrained optimisation, optimisation-based control, gradient-free algorithms (Jeremy)
- Term 2:
 - **Week 7: Test (Friday 24 April 19:00 – 20:00)**
 - Applications (reinforcement learning, power systems optimisation, etc.)
 - Two weeks taught by Prof. Roy Smith on a variety of interesting topics!
- Term 2: Applications

Lecture plan

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Weeks 2-6 are quite mathematical (some would say difficult!).

Advice: go to lectures, ask questions

- Term 2: Applications

Lectures

- Monday 4-5pm E16
- Tuesday 11am-12pm Psych-Soc 115
- Wednesday 11am-12pm Beatrice Tinsley 111 (will sometimes be a tutorial instead of a lecture)

Assessments

Assessment	Weight	Due Date
Test	20%	Week 7, 24 April 2025, 7pm
Project - Progress Report	10%	Week 5, 20 March 2025, 5:00pm
Project - Oral Presentation	10%	Week 12 (regular lecture slots)
Project - Final Report	30%	Week 12, 29 May 2025, 5:00pm
Final Exam	30%	See UC exam notices

Term Test

- Covers weeks 1-6 material only
- 60 minutes
- Week 7, 24 April 2025, 7pm
- Room TBC

New for 2026: The examiners will award a failing grade to students who score less than 40% for the Test and Exam combined.

Projects

- You can do the project as an individual or in a group of 2 people.
- Each lecturer is offering at least one project – pick one soon as the progress report is due Week 5!
- Alternatively, if there's something else you want to do – contact any of us and get approval + agreement to oversee the project.
- More details on LEARN.

New for ENEL courses in 2026

- All deadlines are 5:00pm on the day.
- The late penalty is 10% in absolute terms per day or part of a day.
- For example, an assignment with a nominal mark of 83% submitted 0-24 hours late will receive a mark of 73%, and submitted 24-48 hours late will receive 63%.



AI usage

Assessment Item	Permitted use of AI	
Project presentation	Generative AI Tools Are Permitted for Certain Parts of This Assessment	
Project reports	Generative AI Tools Are Permitted for Certain Parts of This Assessment	
Test	Generative AI tools cannot be used for this assessment.	
Exam	Generative AI tools cannot be used for this assessment.	

In the project-based assessments, you are permitted to use generative artificial intelligence (AI) for the purpose of proof reading and editing documents, for debugging code, and for gathering and summarising knowledge. No other use of generative AI is permitted, including the generation of substantial parts of your computer code or report.

Please include a Statement of AI use (if no AI tool has been used, then this must also be stated) and a listing of all prompts provided to the AI tool, clearly indicating which AI tools were used and how they contributed to your assessment.

Unforeseen circumstances

- In the case of a re-sit of either the Tests or the Exam, the grade in the **second attempt only** is used to calculate the final grade.
- The Exam resit will take place in the Week of Monday 6 July – Thursday 9 July.
- Students should inform the Course Coordinator if they have applied for Special Consideration for the Tests or Exam.
- For project assignments, apply to the Course Coordinator. The usual remedy is an extension.

Appeals, reconsiderations

- Within five working days of receiving the results of, or feedback for, an assessment (whichever is later), a student may apply to the Course Coordinator to have the assessment reconsidered. [Taught and Project Course Assessment Regulation 7(a) [here](#)]
- Within four weeks of the final course grades being released, a student may apply for reconsideration of a final grade on the appropriate form. Only assessment items that have not already been returned to students will be fully re-marked, while the marks of previous assessments will be checked. Note that a fee is charged, but this is refunded if a change of grade occurs. [Taught and Project Course Assessment Regulation 7(d) [here](#)]

Good luck and have fun!!



Optimized route

